W4479 – Econometrics, summer semester 2025

Applied Econometrics Project with R

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Project beginning:	June 10, 2025
Project due:	July 09, 2025, 11:59 p.m.

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Problem 1

a)

Sample Size	Error Distribution	Intercept Mean	Intercept Variance	Slope Mean	Slope Variance
20	Chi-squared	2.7776689	6.6155607	0.8494	0.0008535
80	Chi-squared	2.7914822	1.5191742	0.8494	0.0001896
320	Chi-squared	2.8064968	0.4283770	0.8502	0.0000539
20	Normal	2.3827648	7.0476234	8.5311	0.0008066
80	Normal	2.5069807	1.6306922	0.8517	0.0002017
320	Normal	2.5341764	0.3782726	0.8500	0.0000493

Comments

Intercept: The mean intercept estimates for the normal distribution deviates/less than the true intercept but approaches the true value of 2.75 as the sample sizes increases. The variance decreases significantly as sample size increases across the error distributions.

Slope: The mean slope estimates are very close and approaches the true value of 0.85, and the variance reduces significantly with larger sample sizes for both distributions.

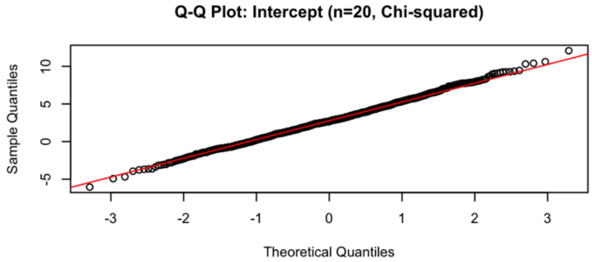
b) Comments

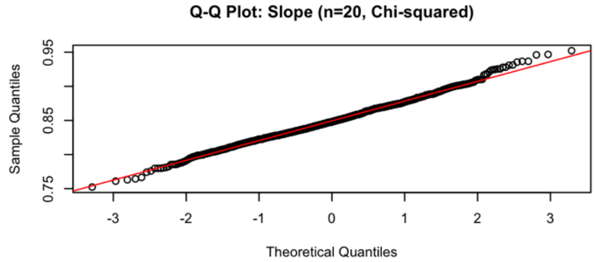
Intercept:

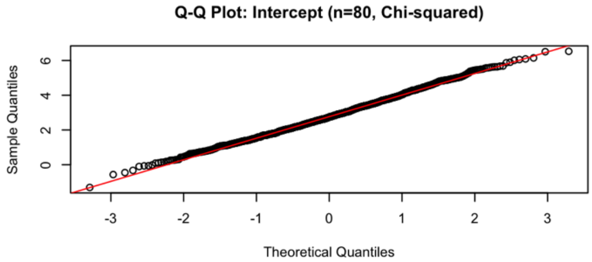
- For **n=20**, the Q-Q plots shows some deviation and the extreme ends of the diagonal line, for both error distributions depicting some non-normality.
- For larger sample sizes **80** and **320** for both error distributions show improved alignment with the diagonal line indicating the estimates are approximately normally distributed, except for normal distribution with sample size 320 which shows significant deviation at the tail.

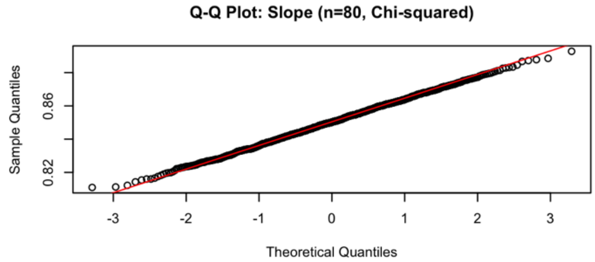
Slope:

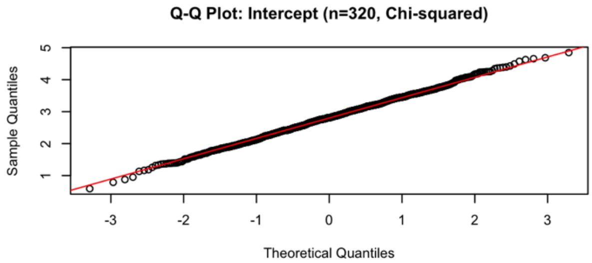
- The Q-Q plot for sample size **20** for **normal** error distribution shows moderate departure at the bottom and tail end of the diagonal line depicting non-normality.
- Whilst remaining plots for the sample size and error distributions shows a good alignment with diagonal line indicating the slope estimates are approximately normally distributed.

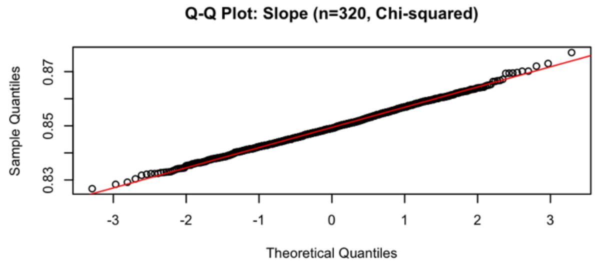

Q-Q Plot: Intercept (n=20, Chi-squared)
Sample Quantiles
Theoretical Quantiles

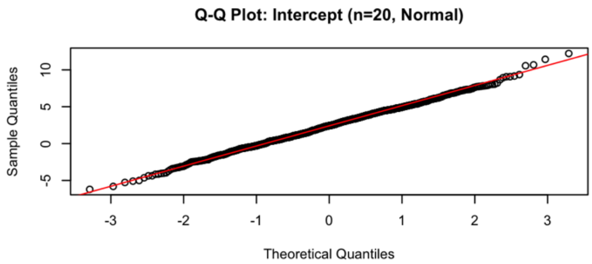

Q-Q Plot: Slope (n=20, Chi-squared)
Sample Quantiles
Theoretical Quantiles

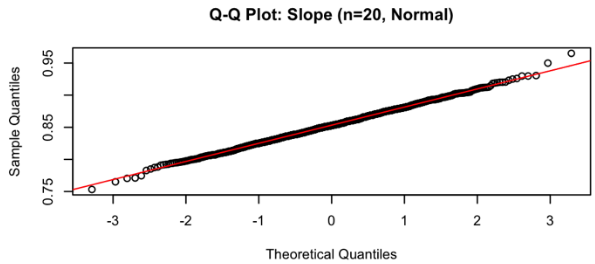

Q-Q Plot: Intercept (n=80, Chi-squared)
Sample Quantiles
Theoretical Quantiles

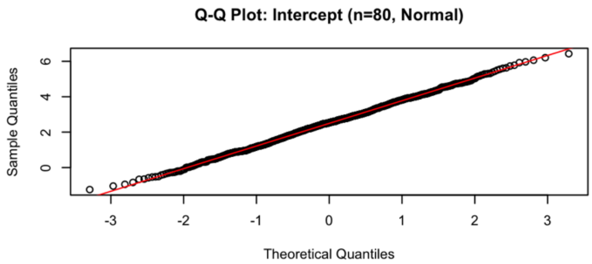

Q-Q Plot: Slope (n=80, Chi-squared)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Intercept (n=320, Chi-squared)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Slope (n=320, Chi-squared)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Intercept (n=20, Normal)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Slope (n=20, Normal)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Intercept (n=80, Normal)
Sample Quantiles
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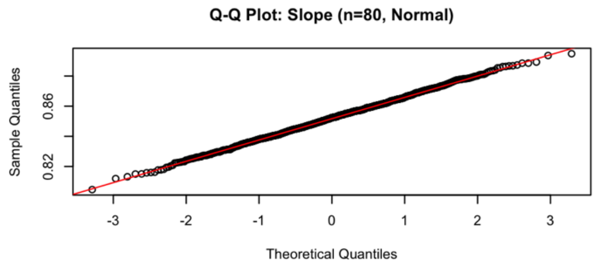
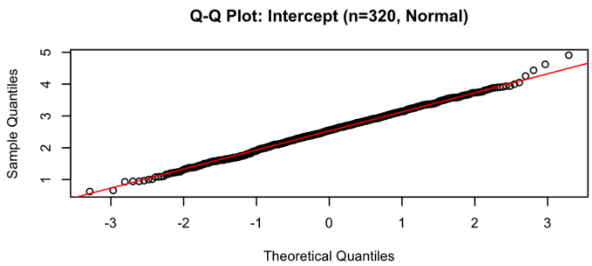
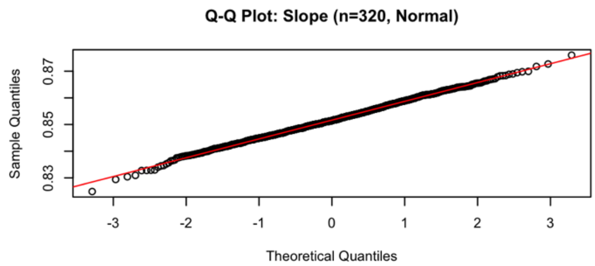

Q-Q Plot: Slope (n=80, Normal)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Intercept (n=320, Normal)
Sample Quantiles
Theoretical Quantiles


Q-Q Plot: Slope (n=320, Normal)
Sample Quantiles
Theoretical Quantiles

c)

Sample Size	Error Distribution	Estimate Type	Test Statistic	P -Value
20	Chi-squared	Intercept	2.61825155	0.2701
20	Chi-squared	Slope	11.28148449	0.0036
80	Chi-squared	Intercept	4.81211035	0.0902
80	Chi-squared	Slope	1.55106646	0.4605
320	Chi-squared	Intercept	0.06490729	0.9681
320	Chi-squared	Slope	2.98690688	0.2246
20	Normal	Intercept	1.34804863	0.5097
20	Normal	Slope	1.42940664	0.4893
80	Normal	Intercept	0.60634529	0.7385
80	Normal	Slope	0.23177852	0.8906
320	Normal	Intercept	0.11868055	0.9424
320	Normal	Slope	0.88561941	0.6422

Comments:

Chi-square with small sample size (n=20) shows significant normality observing ($p < 0.05$) for both intercept and slope. Where as all other combinations shows significant deviations as large p-values a large.

d)

Although the Jarque Bera test shows a significant departure from normality, the empirical results demonstrate that the distribution of OLS estimators depends on sample size, with larger samples reducing variance and enhancing normality.

The Normal error distribution does marginally better estimating the true values for the slope whilst the Chi square error distribution has better estimating the intercept.

The empirical results align well with classical econometric theory, demonstrating normality and diminishing variance as sample size increases, with the observed patterns generally consistent with theoretical expectations.

Problem 2

a) Data Source and Description

Source: The dataset is part of the Woolridge package in R, called "**wage1**". It contains data from the 1976 Current Population Survey.

Dependent Variable (Y) - Wage: Average hourly earnings.

Independent Variable (X) - Education: Years of education.

Variable	Observations	Minimum	Maximum	Mean	Variance
Wage	526	0.53	24.98	5.896103	13.638884
Education	526	0.00	18.00	12.562738	7.667485

Comments:

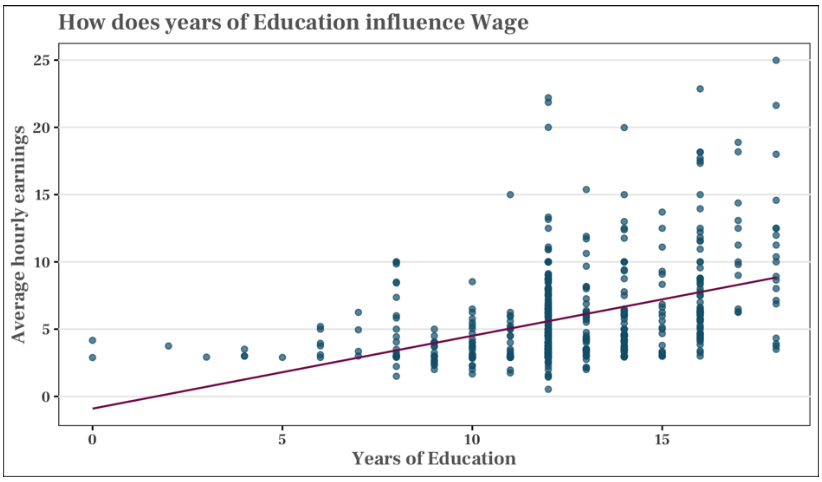
The large range and high variance of the wage variable reflect the data is skewed to the left. Education has a wide range with mean closer to the upper range also showing a right skewed data.
Also, the data has 526 observations for both variables to meet the requirement.

b) Estimated regression equation

$$\hat{Y} = -0.90485 + 0.54136.X$$

Comment on the Visual Fit

The fitted line clearly slopes upwards, indicating that as years of education increase, average hourly earnings tend to increase. This suggesting a moderate linear relationship between Years of Education and Average Hourly Earnings.



c) Logical interpretation

- **Slope of 0.54136:** The positive slope logically indicates that each additional year of education increases average hourly earnings by about **$54**, reflecting a modest but significant return to education.

- **Intercept of -0.90485:** This suggests that without any years of education the average hourly earnings is **$-0.9**. Although, this negative value does not practically make economic meaning.

- The plot shows increasing dispersion at higher years of education, yet the linear model assumes a constant return across all education levels.

d)

- **Statistical Significance of β_2:**
  Estimated slope (β_s) = 0.54136, p-value **< 2e-16**.
  The p-value is below 0.01, so β_2 is statistically significant at the 1% level, indicating years of education significantly affect wage.

- **99% Confidence Intervals:**
  - **Intercept (β_1)**: [-2.6756608, 0.8659575]
  - **Slope (β_2):** [0.4037001, 0.6790184]

  The actual intercept and slope both lies within the lower and upper limits of the 99% confidence interval.

e)

A one-tailed hypothesis test to assess whether the slope coefficient β_2 (the effect of years of education, on average hourly earnings) is positive, reflecting the economic expectation that education increases earnings.

Justification

The scatter plot shows a clear upward trend, suggesting that more years of education are associated with higher average hourly earnings. A one-tailed test is appropriate because the research question focuses on detecting a positive effect, as a negative effect is not economically meaningful in this context.

The test will be performed at a 5% significance level (α=0.05).

Null Hypothesis (H_0): $\beta_2 \leq 0$

Alternative Hypothesis (H_1): $\beta_2 > 0$.

Estimated β_2	t value	*df*	z-value
0.54136	10.167	542	1.645

Decision Rule: Reject H_0 if the t-value > 1.647 (one-tailed critical value at 5% significance).

Decision: The t-value (10.167) is much greater than the critical t-value (1.647), hence, reject the null hypothesis H_0 at the 5% significance level.

Problem 3

a) Data Source and Description

Source: The dataset is produced from US Federal Reserve Bank. It contains US economic time series data available from https://fred.stlouisfed.org.

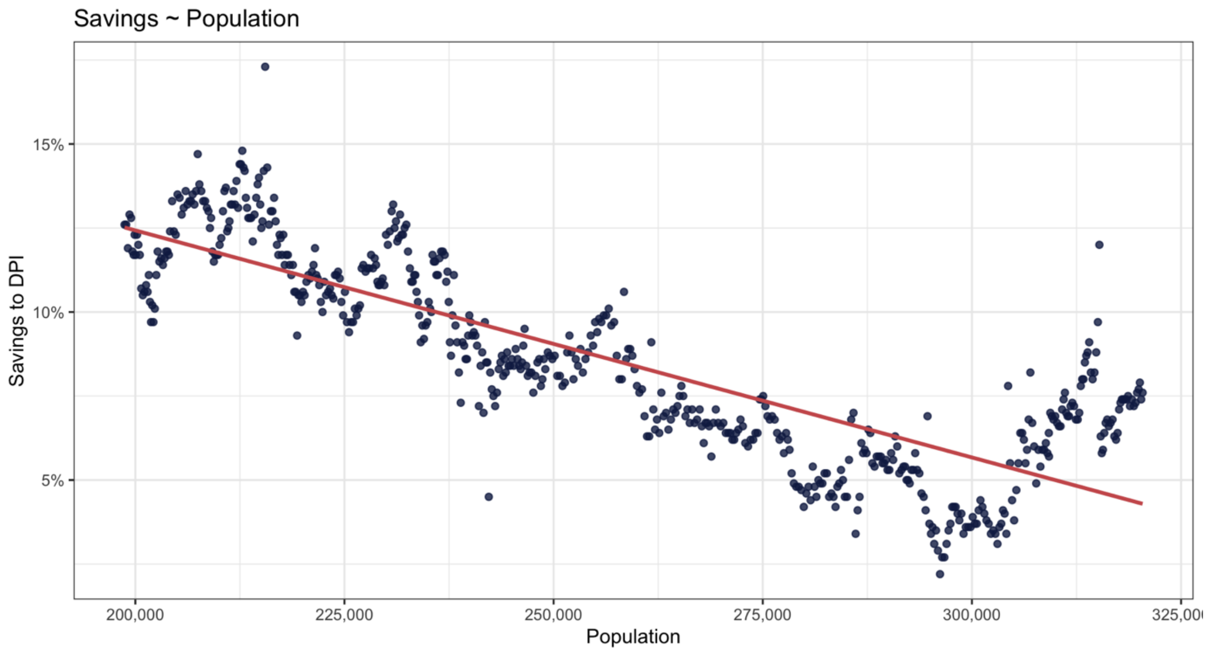
Dependent Variable (Y) - Savings: Personal saving as a percentage of disposable personal income.

Independent Variable (X) - Population: Total population, in thousands.

Also, the data has 574 observations for both variables to meet the requirement.

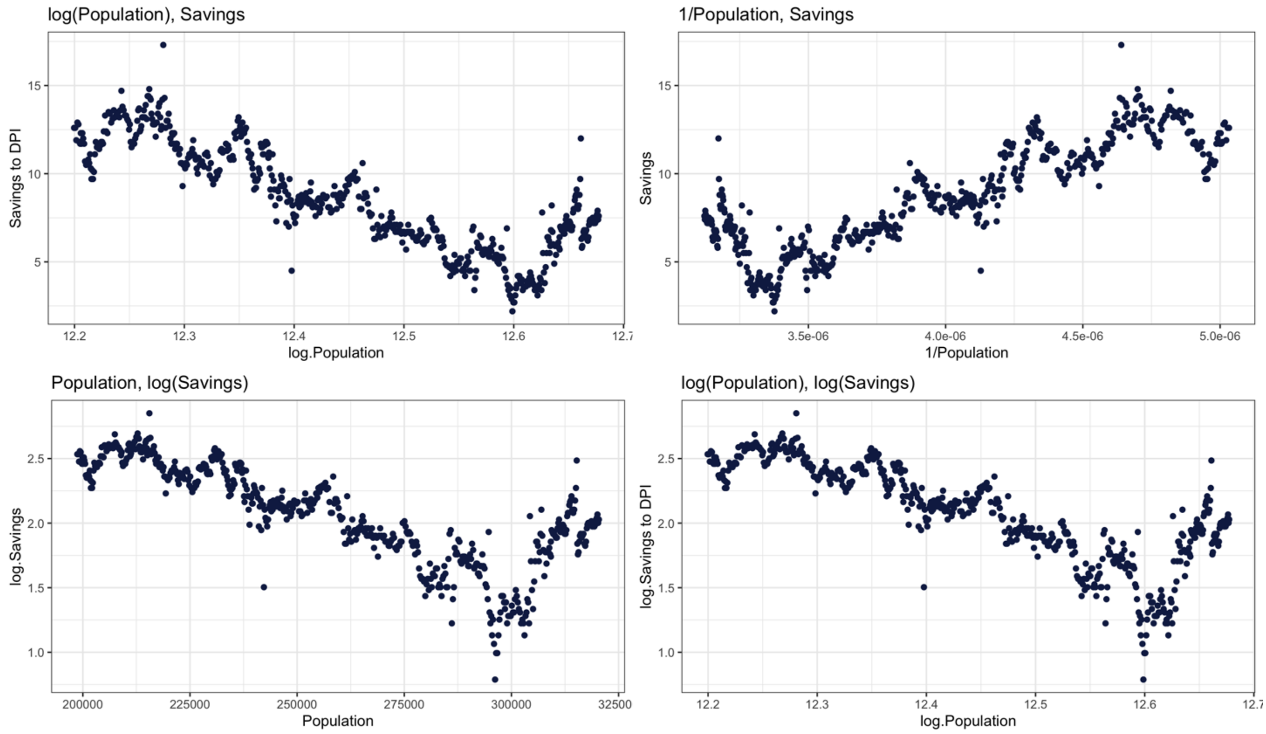
Variable	Observations	Minimum	Maximum	Mean	Variance
Savings	574	2.2	17.3	8.567247	8.786360
Population	574	198,712	320,402.3	257,159.7	1,345,598,000

b)



- **Nonlinear Pattern:**
  As seen in the plot, the relationship between **population** and **unemployment** is **clearly nonlinear**. An SLR assumes a linear relationship and SLR model would fit a straight line through a pattern that bends and curves, leading to poor fit and misleading results.
- As seen in the plot, the relationship between population and savings is clearly nonlinear. An SLR assumes a linear relationship of the form:

c)



Model	Comment
log (Population) vs Savings	Not strongly linear; some non-linearity remains
1/ Population vs Savings	Not strongly linear; some non-linearity remains
Population vs log (Savings)	Moderate improvement; still a bit of nonlinearity
log (Population) vs log(Savings)	Better improvement; still a bit of nonlinearity

d)

Model	Intercept	Slope(β_1)	Standard error(β_1)	t-values(β_1)	R^2
Savings ~ Population	25.95	-6.758e-05	1.852e-06	-36.48	0.6994
Savings ~ Log (Population)	227.2895	-17.5719	0.4573	-38.42	0.7208
Savings ~ 1/ Population	-9.191	4.474	1.124	39.79	0.7346
log (Savings) ~ Population	4.221	-8.322e-06	2.601e-07	- 31.99	0.6415
log(Savings) ~ log(Population)	28.91333	-2.1557	0.06525	-33.04	0.6561

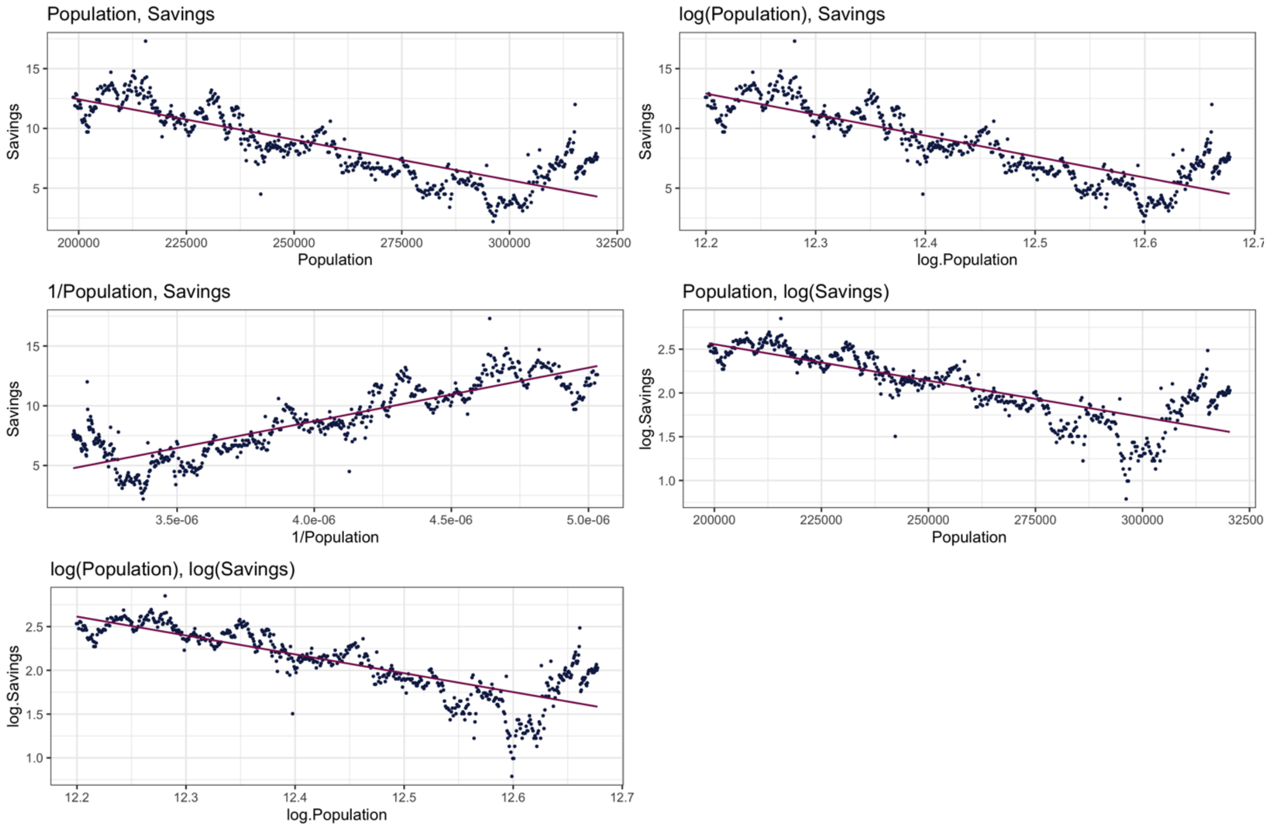
e)

- **log (Savings) ~ Population: log** (Savings) = 4.221 - 8.322e-06 (Population)
- **log (Savings) ~ log (Population):** log (Savings) = 28.91333 - 2.1557 log(Population)

f)

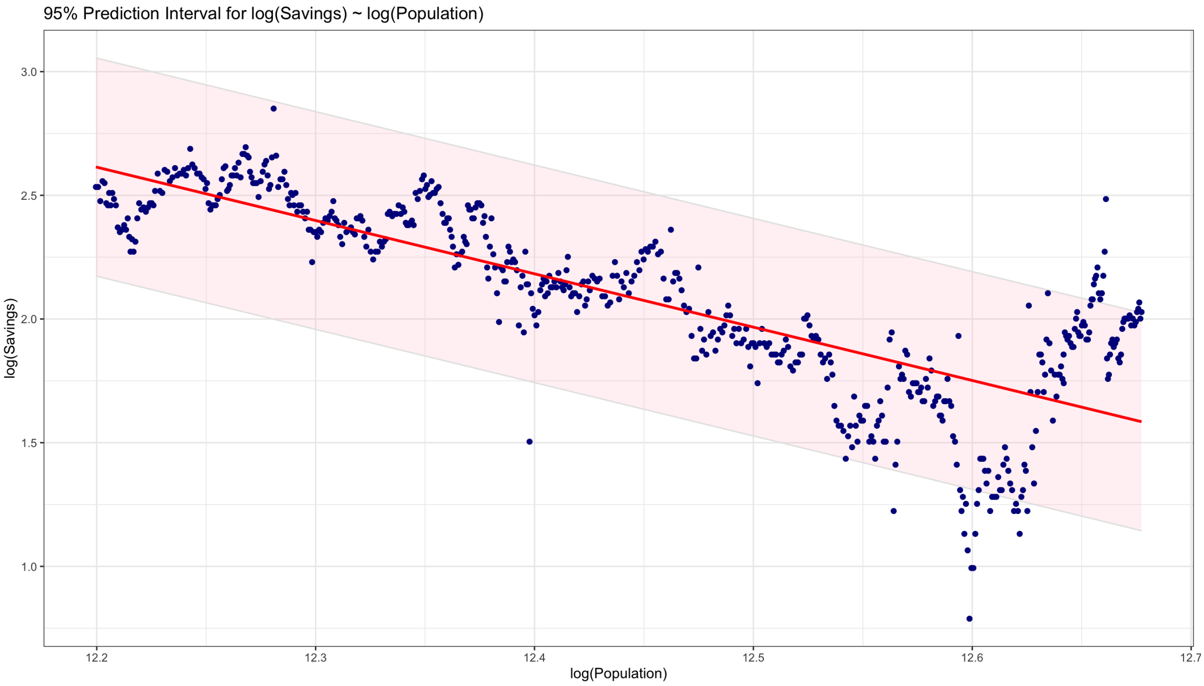
Model	RSS
Population vs Savings	1513.28371
log (Population) vs Savings	1405.80659
1/ Population vs Savings	1336.05861
Population vs log (Savings)	29.84047
log (Population) vs log(Savings)	28.62107

g)



Comment: Yes, there is significant improvement in the preferred model log(Savings) ~ log (Population) compared to the original SLR model. The observed values in the preferred model are much closer to the fitted line, although there is a bit of deviation at the tail of the line.

h)



Problem 4

a) Data Source and Description

Source: The dataset is produced from US Federal Reserve Bank. It contains US economic time series data available from https://fred.stlouisfed.org.

Dependent Variable (Y) - Savings: Personal saving as a percentage of disposable personal income.

Independent Variable (X_2) - Consumption: Personal consumption expenditures, in billions of dollars.

Independent Variable (X_3) - Population: Total population, in thousands.

Independent Variable (X_4) - Unemployed: Total number of unemployed in thousands.

Also, the data has 574 observations for both variables to meet the requirement.

Variable	Observations	Minimum	Maximum	Mean	Variance
Savings	574	2.2	17.3	8.567247	8.786360
Consumption	574	506.7	12193.8	4820.093	12650850
Population	574	198,712	320,402.3	257,159.7	1,345,598,000
Unemployed	574	2685.0	15352.0	7771.31	6979948

b)

Consumption: Negative relationship. Hence, as people spend more, they may save less.

Population: Positive relationship (but ambiguous). A larger population might lead to more aggregate savings, but not necessarily a higher saving rate.

Unemployed: Negative relationship. Higher unemployment may reduce income and hence, savings.

c)

Model	Variable	Intercept(β_0)	Slope(β_i)	p-values(β_0)	p-values(β_i)
Savings ~ Consumption	Consumption	11.750	-0.0006607	0.000	3.91e-125
Savings ~ Population	Population	25.95	-6.758e-05	1.98e-226	1.98e-151
Savings ~ Unemployed	Unemployed	11.26	-0.0003471	2.23e-123	3.38e-14
Savings ~ Cons. + Pop.	Consumption	47.28	0.001078	3.69e-73	1.22e-20
	Population		-0.0001708		4.95e-47
Savings~ Cons.+Unemp	Consumption	9.966	-0.0008071	1.46e-189	1.07e-130
	Unemployed		0.0003206		5.06e-20
Savings ~ Pop + Unemp	Population	27.59	-0.0000865	5.48e-265	7.85e-178
	Unemployed		0.0004136		4.84e-41
Savings ~ Cons. + Pop. + Unemploy	Consumption	51.91	0.001222	5.18e-112	5.55e-37
	Population		-0.0002049		3.81e-84
	Unemployed		0.0004442		2.74e-57

d) Savings ~ Consumption. + Population + Unemployed

The preferred model above has all three predictors significant at extremely low p-values. However, all models have statistically significant coefficients.

e)

Model	**R-squared**	**Adjusted**
Savings ~ Consumption	0.6994	0.6989
Savings ~ Population	0.6286	0.6280
Savings ~ Unemployed	0.0957	0.0941
Savings ~ Cons. + Population	0.7418	0.7409
Savings~ Cons. + Unemployed	0.6794	0.6783
Savings ~ Pop + Unemployed	0.7808	0.7800
Savings ~ Cons. + Pop. + Unemployed	0.8349	0.8340

Savings ~ Consumption. + Population + Unemployed

The preferred model above has the highest R^2 (0.8349) and adjusted R^2. This confirms it is the best-fitting model among the seven.

f)

Variable	**95% CI Lower**	**95% CI Upper**	**99% CI Lower**	**99% CI Upper**
Intercept	48.34	55.48	47.21	56.61
Consumption	0.001047	0.001398	0.000991	0.001454
Population	-0.000222	-0.000188	-0.000228	-0.000182
Unemployed	0.000396	0.000493	0.000380	0.000508

Comment: Yes, there is significant improvement in the preferred model log(Savings) ~ log (Population) compared to the original SLR model. The observed values in the preferred model are much more closer to the fitted line, although there is a bit of deviation at the tail of the line.

g)

Point prediction	**lower**	**upper**
8.567247	5.443205	11.69129